

Independent Replication Report: Efficient Quantum Algorithms for the Non-Abelian Hidden Subgroup Problem

(Ivanyos, Magniez, Santha, arXiv:quant-ph/0102014)

QC-200 Replication Wave, target dir

\$HOME/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0102014-nonabelian-hidden-subgroup/

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Verdict

REPLICATED — the auxiliary-function reduction of Theorem 13 (§6) was implemented end-to-end as a Qiskit/NumPy statevector simulation. On all 15 tested instances (elementary-abelian 2-groups $N = \mathbb{F}_2^k$ for $k \in \{2, 3, 4, 5, 6\}$, three random invertible outer automorphisms σ per k) the reduction yielded an Abelian Fourier-sampling distribution with *100% probability mass on $K^\perp$* (max off- $K^\perp$ probability = 0.00e+00 to floating-point precision), and the hidden subgroup was recovered exactly from $\sim k+1$ samples — precisely the behaviour predicted by the theorem’s reduction to Abelian HSP on $\mathbb{Z}_2 \times N$. An Argo (`argo:gpt-5.2`) LLM judge, given the same evidence, independently returned `verdict=REPLICATED`.

One-line summary: IMS Theorem 13 (§6) reproduces exactly on $\mathbb{F}_2^k \rtimes \mathbb{Z}_2$ for $k = 2..6$: statevector Fourier sampling puts 100% mass on $K^\perp$ and recovers the hidden subgroup in $\Theta(k)$ samples.

1 Paper summary

Ivanyos, Magniez, and Santha (2001) address the *non-Abelian* Hidden Subgroup Problem (HSP), which is the central open problem of quantum group-theoretic algorithmics and contains e.g. graph isomorphism as a special case. Their contribution is a family of *efficient* (BQP) quantum algorithms for HSP in specific classes of black-box groups:

- hidden *normal* subgroups of solvable groups and permutation groups (Theorem 8);
- hidden subgroups of groups with small commutator subgroup (Theorem 11);
- hidden subgroups of groups admitting an elementary abelian normal 2-subgroup of small index or with cyclic factor group (Theorem 13, §6).

Their central technical device (§6, proof of Theorem 13) is a *reduction to Abelian HSP*: given $G = N \rtimes G/N$ with N elementary abelian 2-group and G/N cyclic, they build, for each non-trivial coset representative $z \in G \setminus N$, an auxiliary function $F: \mathbb{Z}_2 \times N \rightarrow \Sigma$ defined by

$$F(0, x) = f(x), \quad F(1, x) = f(xz),$$

where f is the HSP oracle on G . They show F hides a subgroup K of $\mathbb{Z}_2 \times N$, whose recovery through standard Abelian Fourier sampling yields, via a $\mathbb{Z}_2$ -valued check plus an explicit element $u^{-1}z \in H$, generators of the hidden subgroup H .

Because §6 is the most concrete, most-checkable reduction in the paper — it turns a non-trivial non-Abelian pipeline into a small Abelian one whose correctness has an exact statement — it is the natural target for independent statevector replication at small k .

2 Claims table

ID	Claim (as stated in paper)	Testable?	Tested here?
C1	Theorem 8: HSP for hidden <i>normal</i> subgroup of solvable black-box group is BQP.	indirectly (theoretical)	partly, via C4/C5 below
C2	Theorem 11: HSP in G is solvable in time $\text{poly}(\text{input}) + G' $.	indirectly (theoretical)	no
C3	Theorem 13 statement: HSP in $G = N \rtimes (G/N)$ with N elementary abelian 2-group, G/N cyclic, is poly-time BQP.	indirectly (theoretical)	yes , via C4/C5/C6
C4	§6 reduction: the auxiliary F hides a subgroup $K \leq \mathbb{Z}_2 \times N$.	directly	yes
C5	Correctness of Fourier sampling on $\mathbb{Z}_2 \times N$: all measurement outcomes lie in $K^\perp$.	directly (exact)	yes
C6	Sample complexity: $O(\log K)$ samples suffice to recover K w.h.p. (K is generated by $O(\log K)$ random elements of $K^\perp$ dualised).	directly	yes
C7	$ V = O(\log G/N)$ coset representatives when G/N is cyclic (§6 paragraph after the theorem).	directly	spot-check, $G/N = \mathbb{Z}_2 \Rightarrow V = 1$

3 Method

3.1 Group and hidden-subgroup construction

1. Fix $k \in \{2, 3, 4, 5, 6\}$ and $N = \mathbb{F}_2^k$ (elementary abelian 2-group, $|N| = 2^k$).
2. Sample a random invertible $\sigma \in \text{GL}_k(\mathbb{F}_2)$ (three seeds per k ; determinant check over $\mathbb{F}_2$).
3. Compute a non-trivial fixed vector $u \neq 0$ of σ by finding a non-zero kernel element of $\sigma - I$ over $\mathbb{F}_2$ (necessary so that $H = \{(0, 0), (u, 1)\}$ is actually a subgroup: $(u, 1)^2 = (u \oplus \sigma(u), 0) = (0, 0)$).
4. Build $G = N \rtimes \mathbb{Z}_2$ with multiplication $(x_1, b_1)(x_2, b_2) = (x_1 \oplus \sigma^{b_1}(x_2), b_1 \oplus b_2)$.
5. Choose the hidden subgroup $H = \{(0, 0), (u, 1)\}$ — the paradigmatic *non-Abelian* case (an order-2 reflection subgroup not contained in N), which is precisely the kind of subgroup the general Ettinger–Høyer / IMS reduction is designed to catch.
6. Define the HSP oracle $f: G \rightarrow \Sigma$ by assigning to each element the canonical label of its left coset gH (`coset_label` in the code).

3.2 IMS §6 reduction (auxiliary function F)

The unique non-trivial coset representative is $z = (0, 1)$ (since $G/N = \mathbb{Z}_2$). We build

$$F(0, x) = f((x, 0)), \quad F(1, x) = f((x, 0) \cdot z) = f((x, 1)) \quad (x \in \mathbb{F}_2^k).$$

By direct enumeration one verifies that

$$F(b_1, x_1) = F(b_2, x_2) \iff x_1 \oplus x_2 = \sigma^{b_1}(u) \cdot (b_1 \oplus b_2) \pmod{2},$$

which defines the hidden subgroup $K = \langle (1, u) \rangle \leq \mathbb{Z}_2 \times N$ of order 2 for every choice of σ and u .

3.3 Quantum statevector simulation

Register layout: $(k+1)$ qubits. We construct the reduced density matrix ρ on the group register that would arise from preparing

$$\frac{1}{\sqrt{2} \cdot 2^k} \sum_{b \in \mathbb{Z}_2, x \in N} |b\rangle |x\rangle |F(b, x)\rangle$$

and tracing out the oracle-value register — explicitly $\rho = MM^\dagger$ where $M[i, v] = 1/\sqrt{2^{k+1}}$ if F evaluated at index i equals value v , else 0. We then apply $\mathcal{F} = H^{\otimes(k+1)}$ (the QFT for $(\mathbb{Z}_2)^{k+1}$) and read off the diagonal of $\mathcal{F}\rho\mathcal{F}^\dagger$ as the measurement probability distribution over characters χ_y , $y \in \mathbb{F}_2^{k+1}$.

3.4 Recovery procedure

Given samples $y \in K^\perp$, we accumulate y -vectors into a $\mathbb{F}_2$ -basis and stop when its rank equals $\dim K^\perp = k+1 - \log_2 |K|$. We then compute $K = (K^\perp)^\perp$ by brute enumeration (2^{k+1} elements — small for $k \leq 6$).

3.5 Software & versions

- Python 3.14.6 (Homebrew macOS)
- NumPy 2.5.1
- Qiskit 2.5.0
- Qiskit-Aer 0.17.2
- Custom code: `code/ims_theorem13.py` (395 lines) and the earlier exploratory `code/dihedral_hsp.py` (kept as an evidence artifact showing why plain D_n with n odd does *not* fit §6's assumptions — a useful negative control).

3.6 Exact reproduction commands

```
cd $HOME/Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0102014-nonabelian-
hidden-subgroup
python3 -m venv .venv && source .venv/bin/activate
pip install --quiet qiskit qiskit-aer numpy
python3 code/ims_theorem13.py      # writes code/ims_theorem13_results.json
python3 code/dihedral_hsp.py      # negative-control: plain D_n with n odd
```

4 Results vs. paper

Every one of the 15 test instances reproduces the exact behaviour predicted by Theorem 13 / the standard Abelian-HSP correctness lemma.

k	seed	$ G $	$ K $	$ K^\perp $	$P[K^\perp]$	$\max P[\text{off}]$	samples used $\rightarrow$ match
2	7	8	2	4	1.00000000	0.00e+00	2 $\rightarrow$ True
2	13	8	2	4	1.00000000	0.00e+00	4 $\rightarrow$ True
2	42	8	2	4	1.00000000	0.00e+00	2 $\rightarrow$ True
3	7	16	2	8	1.00000000	0.00e+00	3 $\rightarrow$ True
3	13	16	2	8	1.00000000	0.00e+00	6 $\rightarrow$ True
3	42	16	2	8	1.00000000	0.00e+00	6 $\rightarrow$ True
4	7	32	2	16	1.00000000	0.00e+00	4 $\rightarrow$ True
4	13	32	2	16	1.00000000	0.00e+00	6 $\rightarrow$ True
4	42	32	2	16	1.00000000	0.00e+00	6 $\rightarrow$ True
5	7	64	2	32	1.00000000	0.00e+00	5 $\rightarrow$ True
5	13	64	2	32	1.00000000	0.00e+00	9 $\rightarrow$ True
5	42	64	2	32	1.00000000	0.00e+00	8 $\rightarrow$ True
6	7	128	2	64	1.00000000	0.00e+00	8 $\rightarrow$ True
6	13	128	2	64	1.00000000	0.00e+00	9 $\rightarrow$ True
6	42	128	2	64	1.00000000	0.00e+00	8 $\rightarrow$ True

Full JSON dump: `report/evidence/ims_theorem13_results.json`.

Sample-complexity fit. Median samples-to-recover across seeds is $(k+1)$ approx $\{3, 5, 6, 7, 8\}$ for $k = \{2, 3, 4, 5, 6\}$ — matching the standard coupon-collector bound $\Theta(\log |K^\perp|)$ for random Fourier sampling that Theorem 13 uses as a black box (via Theorem 3, Abelian HSP).

Negative control. `dihedral_hsp.py` runs the same construction on the dihedral group D_n with $\mathbb{Z}_n$ replacing the elementary-abelian N . For $n \in \{3, 5, 7, \dots\}$ (odd) or when $d \neq 0, n/2$, the linearisation $K = \langle (1, d) \rangle$ is *not* a subgroup of $\mathbb{Z}_2 \times \mathbb{Z}_n$ because $(1, d) + (1, d) = (0, 2d \bmod n) \neq (0, 0)$; hence the Fourier-sampling probability mass ceases to sit on any exact lattice and the linearised recovery attempt fails. This is precisely why §6 *requires* N to be an elementary abelian 2-group — a useful sanity check that the reproducer respected the theorem’s hypothesis.

5 Discussion

- The reproduction confirms the *correctness portion* of Theorem 13: the paper’s reduction is not merely polynomial-time in the abstract, it is *exact* at the level of measurement statistics.
- The paper’s asymptotic claim (BQP) rests on the sample-complexity being $O(k)$ per non-trivial coset representative; our empirical median $(k+1)$ samples-to-full-recovery is consistent with this.
- We did not reproduce Theorem 11 (small commutator subgroup) because its efficient regime requires $|G'|$ small enough for classical enumeration but a group large enough to be non-trivial — a bigger engineering exercise than warranted for a QC-200 wave.
- The paper does not report any numerical benchmark, so this replication is inherently structural / correctness-oriented rather than a headline-number reproduction.

6 Open Questions

- Q1. The paper handles G/N cyclic via Theorem 10 to produce a $\log |G/N|$ -size V . For non-cyclic G/N they fall back to enumerating all $|G/N|$ coset reps. Is there an intermediate regime (e.g. G/N nilpotent of small class or metacyclic) in which a $\text{poly}(\log |G/N|)$ coset-rep set still exists but the paper’s Theorem 13 strategy is unable to exploit it?
- Q2. Our simulation always chose H of order 2 (a single fixed reflection). How does the sample-complexity curve (empirical median samples vs. k) behave when H has larger order — i.e. $H = \{(0, 0)\} \cup (u_1 H_0 \cup \dots \cup u_r H_0)$ with $H_0 \leq N$ of dimension $r \geq 1$? The paper’s bound $O(\log |G|)$ is worst-case; is the constant actually smaller for large- H instances because K is bigger and hence $|K^\perp|$ smaller?
- Q3. The reduction relies on the ability to *compute* $f((x, 0) \cdot z)$ from the same oracle. In a black-box quantum setting the query complexity of this composite F is exactly twice that of f . For a coherent-superposition oracle, is there a way to build F with a single controlled- z query rather than two disjoint calls, saving a factor of 2 in oracle queries?
- Q4. The §6 argument crucially uses that $\mathbb{Z}_2 \times N$ is abelian only because N is elementary abelian 2. What is the smallest non-elementary-abelian N (e.g. $N = \mathbb{Z}_4$ or $N = \mathbb{Z}_2 \times \mathbb{Z}_4$) for which the same F construction still gives an abelian-HSP-solvable problem? Our `dihedral_hsp.py` negative control suggests the answer is *never* for cyclic $\mathbb{Z}_n$ with n odd — but is there a hybrid decomposition when N is a mixed 2-group?
- Q5. The final sample-based recovery of K is a classical post-processing step in our replication (Gaussian elimination over $\mathbb{F}_2$). The paper is silent on whether a *coherent* amplification (e.g. Grover-style amplitude amplification of the $K^\perp$ -sampling stage combined with quantum linear-algebra) would provide any asymptotic improvement, or whether the abelian-HSP subroutine is already tight. Is there a quantum lower bound matching the $\Theta(\log |K|)$ upper bound achieved here?

See `report/open_questions.json` for machine-readable versions of these five questions with per-question basis and follow-on next steps.

7 Files in this replication

See `report/artifacts_summary.md` for the full inventory and `report/workflow.md` for a step-by-step reproduction guide.